

Prompt Sensitivity and Its Effect on the Stability of LLM-Generated Research Conclusions

Abstract

Large language models (LLMs) are increasingly used in academic workflows for literature discovery, information extraction, evidence synthesis, interpretation, and scientific writing. Although their usefulness depends partly on their ability to follow natural-language instructions, growing evidence shows that their outputs can vary substantially when semantically similar prompts are expressed using different wording, ordering, formatting, or contextual framing. This phenomenon, generally termed **prompt sensitivity**, raises a particularly important problem for research applications: an LLM may generate different research conclusions from essentially the same underlying evidence. Such variation is consequential because scientific conclusions are expected to be reproducible, evidence-grounded, and appropriately calibrated to uncertainty. This paper reviews the literature on prompt sensitivity and examines how it can affect the stability of LLM-generated research conclusions. It synthesizes findings on prompt wording, few-shot example selection and ordering, instruction specificity, reasoning prompts, decoding variability, evaluation methodology, and model-based judging. Recent work suggests that prompt sensitivity is measurable and can be reduced through calibration, prompt ensembles, self-consistency, instruction tuning, and systematic multi-prompt evaluation. However, evidence also indicates that some reported sensitivity may originate from evaluation artifacts rather than genuine changes in semantic understanding. The paper argues that conclusion stability should therefore be treated as a distinct dimension of scientific reliability rather than inferred from single-prompt accuracy. It proposes a research agenda centered on semantic prompt equivalence, conclusion-level stability metrics, evidence-linked evaluation, uncertainty reporting, and human verification. The central implication is that an LLM-generated conclusion should not be regarded as reliable merely because it is fluent, plausible, or correct under one prompt; its robustness across defensible prompt formulations must also be established.

Keywords: large language models; prompt sensitivity; prompt robustness; scientific reasoning; research conclusions; reproducibility; reliability; hallucination; evidence synthesis; LLM evaluation

1. Introduction

Large language models (LLMs) have developed from general-purpose language generators into systems capable of performing increasingly sophisticated reasoning, summarization, information extraction, coding, and question-answering tasks. The emergence of in-context learning demonstrated that models such as GPT-3 can perform tasks from natural-language instructions and a small number of examples without conventional task-specific fine-tuning (Brown et al., 2020). Subsequent work showed that carefully designed prompts can substantially improve performance on reasoning tasks, including mathematical, commonsense, and symbolic problems (Wei et al., 2022). More recent systems have extended these capabilities to professional and academic domains (OpenAI et al., 2023).

These developments have naturally encouraged researchers to use LLMs throughout the research lifecycle. Potential applications include searching and screening literature, extracting information from papers,

synthesizing evidence, drafting manuscripts, generating hypotheses, and interpreting results. Reviews of LLM use in systematic reviews demonstrate that models have already been investigated across many stages of evidence synthesis, although validated applications remain limited (Khraisha et al., 2024; Stevanovic et al., 2025).

A central problem, however, is that LLM behavior is not determined solely by the underlying research question or evidence. It is also affected by how the question is presented. Small changes in prompt wording, formatting, example order, or instruction structure can alter the model's output. Zhao et al. (2021), for example, found that few-shot performance could vary from near chance to near state-of-the-art depending on prompt format, example selection, and example order. Lu et al. (2022) subsequently demonstrated that the ordering of few-shot demonstrations could have a substantial effect on model performance.

This phenomenon is especially significant when an LLM is used to produce a **research conclusion** rather than a simple classification or short answer. A research conclusion typically represents a synthesis of evidence, an interpretation of relationships or effects, and sometimes a statement about implications. If semantically equivalent prompts cause the model to select different evidence, emphasize different findings, change the direction or strength of an interpretation, or express different levels of confidence, then the apparent conclusion may be an artifact of prompt formulation rather than a stable synthesis of the evidence.

The distinction is important. Prompt sensitivity does not necessarily mean that every output is incorrect. A model can produce individually plausible responses while still being unstable across equivalent prompts. Conversely, measured prompt sensitivity may sometimes be exaggerated by evaluation procedures that fail to recognize semantically equivalent answers (Hua et al., 2025). Thus, evaluating LLM reliability requires more than measuring accuracy under one carefully selected prompt.

This paper examines the relationship between prompt sensitivity and the stability of LLM-generated research conclusions. It reviews the conceptual foundations of prompt sensitivity, summarizes major empirical findings, evaluates current mitigation approaches, identifies methodological limitations, and proposes directions for future research. The central thesis is that **conclusion stability should be evaluated explicitly across semantically equivalent prompts and should be treated as a separate reliability dimension from conventional task accuracy**.

2. Background and Related Work

2.1 Large Language Models and Prompt-Based Inference

Modern LLMs are generally transformer-based neural networks trained on large collections of text and optimized primarily for next-token prediction. The scale of pretraining enables them to acquire broad linguistic and statistical representations, while instruction tuning and alignment methods make them more capable of following natural-language requests. GPT-3 demonstrated that sufficiently large models could perform numerous tasks through few-shot prompting without gradient-based parameter updates (Brown et al., 2020). GPT-4 and subsequent models further demonstrated strong performance across academic and

professional benchmarks, although their developers also documented significant limitations in reasoning, factual reliability, and generalization (OpenAI et al., 2023).

Prompting therefore functions as an important interface between the user's intended task and the model's learned representations. A prompt does not merely communicate a question; it can specify task definitions, output formats, examples, reasoning strategies, roles, constraints, and contextual information. Consequently, changes to these components can alter the distribution of generated outputs.

For research applications, this creates an important distinction between **task-level correctness** and **prompt-level stability**. A model might answer one formulation correctly but produce a different answer after a meaning-preserving reformulation. This is particularly concerning when the output is interpreted as a synthesized scientific claim.

2.2 Prompt Sensitivity

Prompt sensitivity can broadly be defined as the degree to which an LLM's output changes when its prompt undergoes a change that is intended to preserve the underlying task or meaning. The perturbation may involve lexical substitutions, changes in syntax, punctuation, formatting, instruction order, example order, or semantic paraphrasing.

Recent research has attempted to operationalize this phenomenon. Chatterjee et al. (2024) introduced POSIX, a Prompt Sensitivity Index based on changes in response log-likelihood under intent-preserving prompt replacements. Zhuo et al. (2024) proposed ProSA and its PromptSensiScore metric, finding that sensitivity varies across models and datasets and that subjective, reasoning-oriented tasks can be particularly sensitive.

Errica et al. (2025) further distinguished **sensitivity** from **consistency** in classification settings. Their framework measures changes in predictions across prompt rephrasings while separately considering whether predictions remain consistent for instances belonging to the same class.

These approaches indicate that prompt sensitivity should not be treated as a binary property. It is better understood as a continuum whose magnitude depends on the model, task, prompt construction, evaluation method, and type of perturbation.

2.3 Few-Shot Example and Ordering Effects

One of the earliest demonstrations of prompt instability concerned few-shot learning. Zhao et al. (2021) found that model performance could vary considerably because of biases introduced by prompt format, example ordering, and label distribution. Their contextual calibration approach reduced these effects by estimating and correcting for biases in the model's output distribution.

Lu et al. (2022) examined example ordering more systematically. Their experiments showed that different permutations of the same few-shot examples could lead to substantially different performance, and that a permutation performing well for one model was not necessarily transferable to another.

For research conclusions, the implications extend beyond classification accuracy. If examples are used to demonstrate how scientific evidence should be weighed or how conclusions should be phrased, their order

and content could influence the model's subsequent interpretation. A model may therefore produce apparently different interpretations even though the underlying evidence remains unchanged.

2.4 Prompting and Reasoning

Chain-of-thought (CoT) prompting provided another important development. Wei et al. (2022) showed that prompting sufficiently large language models to generate intermediate reasoning steps could improve performance on difficult reasoning tasks.

Self-consistency subsequently extended this idea by sampling multiple reasoning paths and selecting the most consistent answer. Wang et al. (2023) reported substantial gains on several reasoning benchmarks using this approach.

These techniques suggest that a single generated reasoning path may not fully represent the model's latent uncertainty. However, consensus among multiple generations should not automatically be equated with truth. If multiple generations share the same underlying misconception or unsupported assumption, aggregation can increase confidence in an incorrect conclusion.

This distinction is especially important for scientific synthesis, where correctness depends on correspondence with external evidence rather than merely internal agreement among generated responses.

3. Prompt Sensitivity and the Stability of Research Conclusions

3.1 What Constitutes a Stable Research Conclusion?

A research conclusion can be considered stable when semantically equivalent, methodologically defensible prompts applied to the same evidence produce conclusions that remain substantially consistent with respect to:

1. **Direction** — whether an effect, association, or relationship is positive, negative, or absent.
2. **Magnitude or qualitative strength** — whether the evidence supports a strong, moderate, weak, or negligible claim.
3. **Scope** — whether the conclusion applies to the population, setting, and conditions actually represented by the evidence.
4. **Causal interpretation** — whether the model distinguishes association from causation.
5. **Evidence selection** — whether the same important findings are considered.
6. **Uncertainty** — whether limitations and confidence are represented consistently.
7. **Implications** — whether practical or theoretical implications remain compatible with the evidence.

This definition is deliberately stricter than textual similarity. Two responses may use completely different wording while communicating the same scientific conclusion. Conversely, two highly similar responses may contain a meaningful difference in causal interpretation or certainty.

3.2 A Conceptual Model

Let E represent a fixed evidence set, Q the research question, P_i a prompt formulation, and M an LLM. The generated conclusion can be represented abstractly as

$$C_i = M(E, Q, P_i, \theta, \epsilon_i),$$

where θ represents the model and inference configuration and ϵ_i represents stochastic generation effects.

For a set of semantically equivalent prompts

$$P = \{P_1, P_2, \dots, P_n\},$$

a stable system should produce conclusions whose scientifically relevant properties are approximately invariant:

$$C_1 \approx C_2 \approx \dots \approx C_n.$$

The approximation cannot reasonably require identical wording. Instead, a conclusion-level similarity function $S(C_i, C_j)$ should compare substantive properties such as evidence claims, effect direction, causal language, uncertainty, and scope.

A useful conceptual measure is therefore

$$CS = 1 - \frac{1}{n(n-1)} \sum_{i \neq j} D(C_i, C_j),$$

where CS denotes conclusion stability and D represents a normalized distance between two conclusions.

Such a metric would differ from conventional prompt-sensitivity measures because it would operate at the **research-claim level** rather than merely comparing token probabilities or exact output strings.

3.3 Sources of Instability

Prompt-induced instability can arise through several mechanisms.

First, lexical framing can change attention. A prompt emphasizing "limitations" may encourage discussion of uncertainty, whereas a semantically related prompt emphasizing "key findings" may produce a more positive synthesis.

Second, task specification can alter the model's interpretation. Underspecified prompts leave more decisions to the model. Recent research suggests that prompt underspecification can substantially increase performance variance and may explain part of the apparent sensitivity attributed to LLMs (Pecher et al., 2026).

Third, prompt structure can influence output probabilities. Liu and Chu (2026) provide a theoretical analysis connecting prompt sensitivity to changes in model gradients and next-token log probabilities. Their analysis further suggests that prompt templates can exert substantial influence on model logits.

Fourth, stochastic decoding introduces an additional source of variability. Even with a fixed prompt, sampling-based generation can yield different outputs. Cui and Alexander (2026), studying LLM-based data analysis, found considerable variation in analytical outcomes across repeated executions even under consistent configurations.

Thus, observed instability should be decomposed into at least two components:

$$V_{\text{total}} = V_{\text{prompt}} + V_{\text{generation}} + V_{\text{interaction}},$$

where the final term captures interactions between prompt formulation and stochastic generation.

4. Why Instability Matters for Scientific Research

4.1 Evidence Synthesis

Scientific research depends heavily on synthesis. A researcher may provide an LLM with abstracts, extracted study results, or a collection of papers and ask it to determine the overall evidence. If alternative prompts cause the system to prioritize different studies or interpretations, the resulting synthesis can become unstable.

This problem is already visible in empirical research on LLM-assisted systematic reviews. Khraisha et al. (2024) found that GPT-4 could approach human-like performance in particular screening conditions, but performance was considerably weaker in other settings and was sensitive to methodological conditions and prompt quality.

A 2025 scoping review similarly found growing applications of LLMs across systematic-review workflows but concluded that fully established and validated applications remained uncommon.

These findings imply that a high-performing single prompt should not be interpreted as evidence that an LLM can reliably perform evidence synthesis in general.

4.2 Scientific Citation and Factuality

Research conclusions depend on accurate references. Hallucinated or incorrectly attributed evidence can therefore undermine an otherwise plausible conclusion.

A comparative study of ChatGPT and Bard in systematic-review reference generation found substantial hallucination and low precision/recall in the tested literature-retrieval tasks. GPT-4 performed better than GPT-3.5 on several measures but still produced substantial errors (Athalye et al., 2024).

The problem is especially dangerous because fluent prose can make unsupported claims appear authoritative. Prompt sensitivity can compound this problem: different prompt formulations may cause a model to retrieve, mention, or emphasize different sources, thereby changing the apparent evidentiary basis of the conclusion.

4.3 Interpretation Versus Retrieval

A particularly important distinction is between **retrieving evidence** and **interpreting evidence**. Prompt sensitivity may be relatively harmless when alternative outputs differ only in wording but remain factually equivalent. It becomes more serious when prompt changes alter interpretation.

For example, consider evidence showing a statistical association between variable X and outcome Y . One prompt may lead the model to conclude:

"The studies indicate an association between X and Y ."

A different prompt might produce:

" X contributes to Y ."

The second statement can imply causality even if the underlying studies were observational. The surface-level difference is small, but the epistemic difference is substantial.

Consequently, research-oriented evaluation should explicitly distinguish **semantic stability** from **epistemic stability**.

5. Current Approaches and Developments

5.1 Prompt Calibration

Contextual calibration is one established strategy for reducing prompt-related biases. Zhao et al. (2021) estimated model preferences using content-free inputs and adjusted output probabilities accordingly. Their results showed improved performance and stability across prompt choices.

Calibration is attractive because it addresses systematic output biases without retraining the entire model. However, its applicability to open-ended scientific conclusions is less straightforward because there is no fixed label set over which probabilities can simply be calibrated.

Future adaptations could instead calibrate dimensions such as evidence selection, confidence, causal language, or claim strength.

5.2 Multi-Prompt Evaluation

Rather than evaluating an LLM using a single prompt, researchers can generate several semantically equivalent formulations and aggregate the resulting outputs.

Multi-prompt evaluation is increasingly recognized as an important methodological principle. The TACL paper *State of What Art? A Call for Multi-Prompt LLM Evaluation* argues that benchmark results can depend on prompt templates and that evaluation should account for this variation rather than treating one template as definitive.

SCORE similarly proposes systematic evaluation of consistency and robustness, reporting that relatively small prompt modifications can produce meaningful accuracy differences on established benchmarks.

For research conclusions, a practical workflow could involve:

1. generating multiple meaning-preserving prompts;
2. obtaining independent conclusions;
3. extracting substantive claims;
4. identifying disagreements;
5. verifying disputed claims against the source evidence;
6. reporting the stability distribution.

This transforms prompt sensitivity from an invisible nuisance into an observable quality-control variable.

5.3 Self-Consistency and Ensemble Reasoning

Self-consistency uses multiple reasoning paths and aggregates their answers (Wang et al., 2023).

For scientific synthesis, a related strategy could generate multiple independent evidence interpretations and identify conclusions supported across a large proportion of them. Such an ensemble could be especially useful when the model is uncertain about the interpretation of complex evidence.

However, consensus does not guarantee correctness. If all outputs inherit a common hallucination or biased premise, self-consistency may simply make the error appear more robust.

Therefore, self-consistency should ideally be combined with external evidence verification rather than used alone.

5.4 Prompt Robustness Benchmarks

PromptBench provides a unified infrastructure for evaluating LLMs under different prompt construction and perturbation conditions. Its purpose is to facilitate systematic evaluation rather than relying on isolated prompts.

Other recent frameworks have expanded the evaluation space. POSIX quantifies prompt sensitivity through likelihood changes, while ProSA examines instance-level sensitivity and the relationship between model confidence and robustness (Chatterjee et al., 2024; Zhuo et al., 2024).

These developments are important foundations for research-conclusion stability evaluation, but existing metrics often focus on classification accuracy, likelihood, or general task performance rather than scientific claim stability.

5.5 Instruction Tuning and Robustness Training

An alternative to inference-time prompt engineering is to modify model training. Yan et al. (2024), for example, proposed Contrastive Instruction Tuning, which encourages similar representations for semantically equivalent instruction-instance pairs. Their experiments on PromptBench reported improved robustness to several levels of instruction variation.

More recent research has continued toward robustness-aware training. Such approaches suggest that prompt sensitivity can potentially be addressed at the model level rather than requiring users to discover an optimal prompt.

However, robustness training can introduce trade-offs. A model optimized to produce invariant outputs might become insufficiently sensitive to legitimate changes in task requirements. The goal should therefore be **semantic invariance**, not indiscriminate invariance.

6. Evaluation: From Prompt Robustness to Conclusion Stability

A major methodological gap in existing literature is the absence of a standardized framework connecting prompt sensitivity to the stability of scientific conclusions.

A suitable evaluation framework should operate at multiple levels.

6.1 Prompt-Level Stability

The first level measures whether semantically equivalent prompts produce similar model behavior. Existing metrics such as POSIX and PromptSensiScore can contribute to this layer (Chatterjee et al., 2024; Zhuo et al., 2024).

6.2 Claim-Level Stability

The second level should extract factual and inferential claims from each response.

For example:

- Claim A: treatment T improves outcome Y .
- Claim B: treatment T does not significantly improve outcome Y .
- Claim C: evidence is insufficient to determine an effect.

These statements should not be treated as equivalent merely because their wording overlaps substantially.

6.3 Evidence-Level Stability

Each generated claim should be linked to supporting evidence. A robust conclusion should identify substantially the same relevant evidence regardless of reasonable prompt formulation.

This can be represented using an evidence-support matrix:

$$A_{ij} = \begin{cases} 1 & \text{if conclusion } C_i \text{ is supported by evidence } E_j, \\ 0 & \text{otherwise.} \end{cases}$$

Differences among matrices generated from equivalent prompts can then reveal whether prompt changes alter the evidentiary basis of the conclusion.

6.4 Epistemic Stability

The fourth level concerns uncertainty. A scientifically responsible system should distinguish:

- evidence-supported findings;
- plausible interpretations;
- unresolved questions;
- unsupported claims.

Two outputs may agree on the direction of an effect while differing substantially in certainty. Therefore, stability metrics should assess not only *what* the model concludes but also *how strongly* it concludes it.

6.5 Human and External Verification

LLM-based evaluation alone is insufficient. Evans et al. (2024) found that LLMs can provide useful assessments of scientific synthesis but that their judgments showed weak correlation with human ratings in deeper statistical analysis.

Human experts and independently verifiable source evidence should therefore remain part of high-stakes scientific evaluation.

7. Research Challenges and Limitations

7.1 Defining Semantic Equivalence

The most fundamental methodological challenge is determining whether two prompts are genuinely equivalent.

Paraphrases can unintentionally change scope, emphasis, assumptions, or requested output. A prompt asking for "the strongest evidence supporting the hypothesis" is not equivalent to one asking for "the overall evidence regarding the hypothesis," even though the two may appear similar.

Future benchmark construction must therefore use human validation or formal semantic criteria to establish equivalence.

7.2 Distinguishing Sensitivity from Appropriate Adaptation

Not every output difference is a failure. If a prompt explicitly changes the intended audience, evidence standard, or decision criterion, a corresponding change in output is desirable.

The research challenge is consequently to identify **meaning-preserving perturbations** and distinguish them from legitimate task changes.

7.3 Evaluation Artifacts

Hua et al. (2025) demonstrate that apparent prompt sensitivity can partly arise from evaluation methods such as rigid answer matching or log-likelihood scoring. Their study found reduced performance variance when LLM-based judging was used to recognize semantically equivalent answers.

This finding is important because it prevents an overly simplistic interpretation of robustness results. A model may be semantically stable even when conventional metrics report instability.

7.4 Model and Version Dependence

Prompt sensitivity is not constant across models. Zhuo et al. (2024) reported differences across model sizes and tasks, while Alhetelah and Ahmad (2026) found substantial variation in sensitivity to paraphrased opinion prompts across contemporary models.

Moreover, commercial models can change over time. A prompt that is stable under one model version may behave differently after a model update. Reproducible research therefore requires recording the exact model identifier, inference settings, prompt, date, and relevant system instructions.

7.5 Proprietary Models and Limited Reproducibility

Researchers often lack access to the weights, training data, internal representations, or detailed decoding mechanisms of proprietary LLMs. This makes it difficult to determine why a particular prompt causes a change.

Even if the same API is used, model updates may alter behavior. Consequently, reproducibility requires comprehensive experimental logging and, where possible, frozen model versions.

7.6 Cost and Computational Burden

Multi-prompt and multi-sample evaluation can multiply inference costs. If n prompt variants are tested with k repeated generations each, the number of model calls becomes approximately

$$N = nk.$$

Large-scale scientific evaluation can therefore become expensive, particularly for long-context evidence synthesis.

Efficient sampling strategies are needed to estimate stability without exhaustively testing every possible prompt formulation.

8. Research Gaps and Future Directions

8.1 A Standard Definition of Research-Conclusion Stability

Existing research has produced multiple prompt-sensitivity metrics, but there is no widely adopted standard specifically for **research-conclusion stability**.

Future work should define a benchmark that measures:

- conclusion agreement;
- evidence overlap;
- effect-direction consistency;
- causal-language consistency;
- uncertainty consistency;
- citation consistency; and
- scope consistency.

8.2 Prompt-Equivalence Benchmarks

Benchmark datasets should contain groups of independently validated, semantically equivalent research prompts.

Each group could include variations in:

- wording;
- sentence structure;
- punctuation;
- instruction order;
- formatting;
- explicitness;
- terminology;
- researcher role descriptions.

Models could then be evaluated on whether they preserve the same scientific conclusion across the set.

8.3 Evidence-Grounded Conclusion Evaluation

A particularly promising direction is to move from answer agreement toward **evidence-grounded agreement**.

Suppose five prompts produce the same conclusion. This should not automatically receive a high reliability score. Instead, evaluators should determine whether the conclusion is actually supported by the underlying studies.

A future benchmark could therefore score a system according to:

$$R = f(S_c, E_s, U_c),$$

where:

- S_c = conclusion stability;
- E_s = evidence support;
- U_c = uncertainty calibration.

This would make reliability multidimensional rather than reducible to consistency alone.

8.4 Stability-Aware Scientific Assistants

Future research systems could automatically perform a "prompt robustness check" before presenting a conclusion. The system could generate several semantically equivalent internal prompts, compare conclusions, and warn the user when substantive disagreement occurs.

For example:

Conclusion stability: Moderate.

7/10 prompt variants support the same direction of effect.

3/10 interpret the evidence as insufficient.

Main disagreement: causal versus correlational interpretation.

Such reporting would be more scientifically informative than presenting a single polished answer.

8.5 Integration with Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) offers another potential solution because it grounds responses in explicitly retrieved sources. However, retrieval itself may be prompt-sensitive. Different prompts can produce different search queries or evidence rankings.

Future research should therefore evaluate **end-to-end prompt sensitivity**, including:

Prompt → Retrieval → Evidence Selection → Reasoning → Conclusion.

This pipeline perspective is essential because instability may arise before generation even begins.

8.6 Robustness Without Over-Invariance

The objective should not be to make an LLM produce identical outputs under every prompt. Scientific reasoning requires sensitivity to meaningful distinctions.

The desired property is instead:

semantic equivalence $\Rightarrow$ stable scientific conclusion,

while

semantic difference $\Rightarrow$ appropriately different conclusion.

This principle should guide both robustness training and evaluation.

8.7 Reproducibility Standards

Research papers using LLM-generated conclusions should report at least:

1. model name and version;
2. system prompt;
3. user prompt;
4. prompt variants tested;
5. temperature and decoding parameters;
6. number of repeated generations;
7. retrieved sources;
8. model update date where known;
9. evaluation metrics;
10. human verification procedure.

Such reporting would make LLM-assisted research more reproducible and allow readers to distinguish stable findings from prompt-specific outputs.

9. Conclusion

Prompt sensitivity represents an important but incompletely understood threat to the reliability of LLM-assisted scientific research. Existing studies demonstrate that relatively small changes in prompt wording, example ordering, formatting, and task specification can alter LLM behavior. Research has also shown that some of this variability can be reduced through calibration, multi-prompt evaluation, self-consistency, and robustness-oriented training (Zhao et al., 2021; Wang et al., 2023; Yan et al., 2024).

For scientific applications, however, conventional prompt-robustness metrics are not sufficient. The important question is not simply whether a model's token probabilities or benchmark accuracy change, but whether the **scientific conclusion changes**. A minor lexical difference is inconsequential if the evidence, interpretation, uncertainty, and scope remain stable. Conversely, a small change in wording is consequential if it transforms an association into a causal claim, reverses the direction of an effect, changes the evidentiary basis, or suppresses an important limitation.

Recent work reinforces the need for this distinction. Prompt-sensitivity studies show measurable instability, while newer investigations also demonstrate that some apparent sensitivity can arise from flawed evaluation procedures or underspecified prompts.

The most defensible approach is therefore not to ask whether an LLM is "prompt sensitive" in the abstract, but to determine **how sensitive its scientifically relevant conclusions are to meaning-preserving prompt variations**. Future research should establish standardized prompt-equivalence datasets,

conclusion-level stability metrics, evidence-grounded evaluation protocols, uncertainty measures, and reproducibility standards.

Until such methods become routine, LLM-generated research conclusions should be treated as **candidate interpretations requiring verification rather than autonomous scientific findings**. Prompt robustness is not merely an engineering concern; it is an epistemic property that directly affects the reproducibility, transparency, and trustworthiness of AI-assisted scientific reasoning.

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Note on source integrity. The references above were selected from scholarly publisher pages, ACL Anthology, PMLR, JMLR, ICLR/NeurIPS records, PubMed-indexed literature, and arXiv records. Where a DOI or arXiv identifier was available, it is included rather than inventing one. The literature also contains an important methodological disagreement: some studies report substantial intrinsic prompt sensitivity, whereas newer work suggests that part of the observed effect can result from underspecified prompts or evaluation artifacts. This paper therefore treats prompt sensitivity as an empirical phenomenon whose magnitude must be measured at the level of scientifically meaningful conclusions rather than assumed from benchmark variation alone.